

Linear Algebra: Worked Solutions

Problem 1. The four subspaces of a linear map

Problem. Let $A \in \mathbb{R}^{4 \times 7}$ have rank three. Give the dimensions of its row space, column space, null space, and left null space. State when $Ax = b$ has a solution and whether that solution can be unique.

Solution. The rows of A lie in $\mathbb{R}^7$ and its columns lie in $\mathbb{R}^4$. Row rank and column rank both equal three. Applying rank-nullity to $A : \mathbb{R}^7 \rightarrow \mathbb{R}^4$ and to $A^T : \mathbb{R}^4 \rightarrow \mathbb{R}^7$ gives

Subspace	Ambient space	Dimension
$\text{col}(A^T)$ (row space)	$\mathbb{R}^7$	3
$\text{col}(A)$ (column space)	$\mathbb{R}^4$	3
$\text{null}(A)$	$\mathbb{R}^7$	$7 - 3 = 4$
$\text{null}(A^T)$ (left null space)	$\mathbb{R}^4$	$4 - 3 = 1$

For $b \in \mathbb{R}^4$, the equation is solvable exactly when $b \in \text{col}(A)$. Equivalently, $z^T b = 0$ for every $z \in \text{null}(A^T)$, since $\text{col}(A) = \text{null}(A^T)^\perp$. A nonzero vector spanning the one-dimensional left null space therefore supplies the single independent compatibility condition on b .

If x_0 is a solution, then $A(x_0 + v) = b$ for every $v \in \text{null}(A)$. Conversely, any other solution x satisfies $A(x - x_0) = 0$. Thus the complete solution set is

$$\{x : Ax = b\} = x_0 + \text{null}(A),$$

a four-dimensional affine space. A solvable system cannot have a unique solution because its null space contains nonzero vectors.

Problem 2. Projection onto a line

Problem. For $u = (1, 2)^T$, compute $P = uu^T / (u^T u)$. Verify $P^T = P$ and $P^2 = P$ directly. Explain why its eigenvalues are zero and one and why rescaling u leaves P unchanged.

Solution. Since $u^T u = 5$,

$$P = \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

This matrix is symmetric, and direct multiplication gives

$$P^2 = \frac{1}{25} \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = P.$$

If $Pv = \lambda v$ with $v \neq 0$, idempotence implies $\lambda^2 v = \lambda v$, so $\lambda \in \{0, 1\}$. Both values occur: $Pu = u$ and $P(-2, 1)^T = 0$. These two independent eigenvectors span $\mathbb{R}^2$. The map retains the component along u and discards its orthogonal component.

For any nonzero scalar c ,

$$\frac{(cu)(cu)^T}{(cu)^T(cu)} = \frac{c^2 uu^T}{c^2 u^T u} = P.$$

Rescaling changes the vector used to describe the line, not the line or its projection. The restriction $c \neq 0$ is necessary: the zero vector gives a zero denominator and does not specify a line.

Problem 3. Orthonormal coordinates and contraction

Problem. Suppose $Q \in \mathbb{R}^{m \times d}$ has orthonormal columns. Prove that $\|Qz\|_2 = \|z\|_2$, that $\|Q^T b\|_2 \leq \|b\|_2$, and that equality in the latter holds exactly when $b \in \text{col}(Q)$.

Solution. Orthonormal columns mean $Q^T Q = I_d$. For $z \in \mathbb{R}^d$,

$$\|Qz\|_2^2 = z^T Q^T Q z = z^T z = \|z\|_2^2.$$

Taking nonnegative square roots gives the first claim.

For $b \in \mathbb{R}^m$, write

$$b = QQ^T b + r, \quad r = (I_m - QQ^T)b.$$

Because $Q^T r = Q^T b - Q^T QQ^T b = 0$, the residual is orthogonal to every column of Q , hence to $QQ^T b$. The Pythagorean identity and the first claim, applied to $z = Q^T b$, yield

$$\|b\|_2^2 = \|QQ^T b\|_2^2 + \|r\|_2^2 = \|Q^T b\|_2^2 + \|r\|_2^2.$$

This proves contraction. Equality holds precisely when $r = 0$, or $b = QQ^T b$, which implies $b \in \text{col}(Q)$. Conversely, if $b = Qz$, then $QQ^T b = QQ^T Qz = Qz = b$, so $r = 0$ and equality holds. When $d = m$, the column space is all of $\mathbb{R}^m$, and equality holds for every b .

Problem 4. Singular least squares and minimum norm

Problem. For $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ and $b = (1, 2)^T$, describe all least-squares minimizers and identify the minimum-norm one. Explain why forming $(A^T A)^{-1}$ fails even though the approximation problem has an answer.

Solution. For $x = (x_1, x_2)^T$,

$$Ax = (x_1 + x_2) \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \|Ax - b\|_2^2 = 5(x_1 + x_2 - 1)^2.$$

The objective has minimum zero, attained exactly when $x_1 + x_2 = 1$. Thus all minimizers are

$$x(t) = \begin{bmatrix} t \\ 1 - t \end{bmatrix}, \quad t \in \mathbb{R}.$$

Their squared norms satisfy

$$\|x(t)\|_2^2 = t^2 + (1 - t)^2 = 2(t - \frac{1}{2})^2 + \frac{1}{2}.$$

The unique minimum-norm minimizer is therefore $x^\dagger = (1/2, 1/2)^T$, of norm $1/\sqrt{2}$. Geometrically, every solution is $x^\dagger + s(1, -1)^T$, and $x^\dagger$ is orthogonal to the null direction $(1, -1)^T$. Adding a nonzero null component strictly increases its norm without changing Ax .

The normal-equation matrix and right-hand side are

$$A^T A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 5 \\ 5 \end{bmatrix}.$$

The determinant is zero, so the inverse does not exist. The normal equations still have solutions; they reduce to the same constraint $x_1 + x_2 = 1$. It is the inverse formula that requires full column rank, not the existence of a least-squares minimizer. The pseudoinverse or an SVD calculation selects $x^\dagger$ directly.

Problem 5. Singular values, approximation, and conditioning

Problem. A matrix has singular values 4, 3, 1. Find its operator norm, Frobenius norm, and the optimal rank-one and rank-two Frobenius errors. Give the condition number if it is square and invertible.

Solution. Write an SVD as $A = U\Sigma V^T$, with the listed singular values in decreasing order. Orthogonal changes of coordinates preserve Euclidean lengths, so the largest possible amplification $\|Ax\|_2/\|x\|_2$ is the largest singular value. Therefore

$$\|A\|_2 = 4, \quad \|A\|_F = \sqrt{4^2 + 3^2 + 1^2} = \sqrt{26}.$$

The squared Frobenius norm is the sum of squared singular values because it is invariant under the orthogonal factors.

Retaining the first k terms of the SVD gives the best approximation of rank at most k in Frobenius norm. Its squared error is the sum of squares of the discarded singular values. Here,

$$\min_{\text{rank}(B) \leq 1} \|A - B\|_F = \sqrt{3^2 + 1^2} = \sqrt{10}, \quad \min_{\text{rank}(B) \leq 2} \|A - B\|_F = 1.$$

To see the lower-bound mechanism, project onto the column space of any candidate B . If that space has dimension at most k , projecting A onto it can retain at most the sum of the largest k squared singular values. The orthogonal residual of A is also a component of $A - B$, so its squared norm is a lower bound on the approximation error. The first k left singular vectors attain this bound.

If A is square and invertible with these singular values, then $\|A^{-1}\|_2 = 1/\sigma_{\min} = 1$. Its spectral condition number is

$$\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{4}{1} = 4.$$

Problem 6. Applying a column map to a row batch

Problem. Let $X \in \mathbb{R}^{n \times d}$ contain row observations and let $A \in \mathbb{R}^{m \times d}$. Derive the batch expression for $x \mapsto Ax + b$. State the shape of every factor, including the bias-replication term.

Solution. The i th input is mathematically a column $x_i \in \mathbb{R}^{d \times 1}$, while row i of X is x_i^T . For a column bias $b \in \mathbb{R}^{m \times 1}$, transposing the single-example output gives

$$(Ax_i + b)^T = x_i^T A^T + b^T.$$

Stacking these rows yields

$$Z = XA^T + \mathbf{1}_n b^T \in \mathbb{R}^{n \times m}.$$

The products have shapes

$$\underbrace{X}_{n \times d} \underbrace{A^T}_{d \times m}, \quad \underbrace{\mathbf{1}_n}_{n \times 1} \underbrace{b^T}_{1 \times m}.$$

The outer product $\mathbf{1}_n b^T$ repeats the same bias row n times. With NumPy storage b of shape $(m,)$, the expression `X @ A.T + b[None, :]` performs precisely this replication by broadcasting. A one-dimensional array has no row/column orientation; the added axis explicitly selects the batch broadcasting direction.